PY-STEM · ACTIVITY PYS-07 · OPTICS + LIGHT

Pinhole Precision Challenge

Build the sharpest camera image with no lens.

Win condition: Best image clarity and concept explanation after one aperture redesign.

THE SCIENCE YOU ARE USING

Light travels in straight lines. A pinhole camera has no lens. Light from each point of a scene travels in a straight line through a tiny hole and lands on the screen, forming an upside-down image. Because the rays cross at the hole, the picture is flipped.

The aperture tradeoff. There is a BEST hole size. Too big is bright but blurry. Too tiny is dim and ALSO blurry, because light spreads out at a very small hole. Test a few sizes, find the sharpest, then redesign once.

HOW TO BUILD AND RUN IT

01 **Build the viewer.** Make a tube or box with a foil aperture at one end and a wax-paper screen at the other. You will look THROUGH the wax paper toward the light.

02 **Punch a clean hole.** Pierce a clean, round hole in the foil with a sharp pin and twist it. About pinhead size to start; do not chase a hole that is as tiny as you can make it.

03 **Find the image.** Dim the room lights and aim at a bright source (a flashlight or sunny window). Look THROUGH the wax-paper screen for the upside-down image. A cut-out shape taped over the flashlight makes the flip easy to see.

04 **Tune the aperture.** Try a slightly larger or smaller hole. Balance sharpness against brightness.

05 **Explain the optics.** Describe why the image is flipped and why hole size changes the result.

Safety: No direct sun viewing. Pins handled at a controlled station. Allergy: None.

MATERIALS

Cardstock	shared
Foil	for aperture
Tape	shared
Pushpins or paper clips	shared
Wax paper (for the screen)	per team
Light source	shared

YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

Prediction (what will work best, and why): ______

The ONE variable we are testing: ______

Baseline result: ______

After redesign: ______

DEFEND YOUR DESIGN

What evidence made you change your design? ______

If you ran it again, what is the ONE thing you would change? ______

HOW YOU SCORE (100 POINTS)

SCORING CRITERION	POINTS
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	**100**